

YOGESH KUMAR

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PROFESSIONAL SUMMARY

Full-Stack & Systems Software Engineer with 2+ years of experience engineering scalable, production-grade web systems, event-driven backend microservices, spatial databases, and automated CI/CD pipelines for the Survey Settlement & Land Records Department, Government of Karnataka. Proven expertise in Java 21, Spring Boot, React, Next.js 15, PostgreSQL/PostGIS, GeoServer, and Linux/Docker infrastructure, supporting mission-critical workflows across Karnataka.

TECHNICAL SKILLS

Languages: Java (17/21), JavaScript (ES6+), TypeScript, SQL, Python, C/C++

Backend: Spring Boot, Spring Data JPA, Spring Security, REST APIs, Node.js, Express, LISTEN/NOTIFY

Frontend: React, Next.js (App Router), TypeScript, Tailwind CSS, OpenLayers, Chart.js, Recharts

Databases & Spatial: PostgreSQL, PostGIS, MongoDB, PL/pgSQL, Spatial Indexing (GIST), Query Optimization

GIS Infrastructure: GeoServer (WMS/WFS/WFS-T), QGIS, Spatial Data Pipelines, SLD Symbolology, Java Plugins

DevOps & Systems: Linux (Ubuntu/Daily Driver), Docker, GitLab CI/CD, Jenkins, Nginx, MinIO, Git, Postman

PROFESSIONAL EXPERIENCE

GIS Engineer / Full-Stack Software Development Engineer | May 2023 – Present

Survey Settlement & Land Records Dept, Karnataka (via Invensis Technologies) • Bengaluru, KA

- Architect and maintain high-concurrency production web applications and backend services serving thousands of surveyors statewide.
- Engineered scalable REST APIs & spatial ingestion pipelines in Java 21 & Spring Boot integrated with PostgreSQL/PostGIS.
- Built interactive Next.js 15 / React frontends with OpenLayers spatial mapping, cadastral hierarchy filters, and PDF export reporting.
- Designed schemas, migrations, and PL/pgSQL procedures with GIST indexing for sub-second spatial querying across district datasets.
- Configured and hardened GeoServer instances (WMS/WFS-T layers, SLD styling, and user-context security) for real-time map serving.
- Automated deployment workflows using GitLab CI/CD pipelines, Docker containerization, and Nginx reverse proxies on Linux servers.
- Partnered with department stakeholders to translate administrative requirements into scalable, fault-tolerant technical architectures.

KEY PRODUCTION SYSTEMS

Karnataka State Lake Statistics Platform | [Java 21](#), [Spring Boot](#), [Next.js](#), [PostgreSQL](#), [PostGIS](#), [OpenLayers](#), [GeoServer](#), [Docker](#)

- Engineered state-level platform displaying lake boundaries, encroachment analytics, and survey progress across all Karnataka districts.
- Built Spring Boot APIs for 6-level geographic hierarchy and Next.js frontend with OpenLayers + Google Satellite WMS integration.
- Implemented interactive analytics dashboards and automated PDF abstract generation with embedded maps, sketches, and survey data.

Mojini Land Records Ingestion & Spatial Service | [Spring Boot](#), [PostgreSQL](#), [PostGIS](#), [PL/pgSQL](#), [GeoServer WFS-T](#), [Docker](#)

- Engineered high-throughput pipeline converting non-spatial Bhoomi land survey records into GIS-ready spatial geometry.
- Built event-driven architecture using PostgreSQL LISTEN/NOTIFY and concurrent-safe batching (FOR UPDATE SKIP LOCKED, batch 100).
- Designed 7-level cadastral hierarchy matching algorithm with fallback resolution and automated 50m placeholder geometry generation.

GIS Equipment Lifecycle Management Platform | [Java 21](#), [Spring Boot](#), [Next.js 15](#), [React 19](#), [TypeScript](#), [PostgreSQL](#), [Tailwind CSS](#), [Docker](#)

- Developed enterprise tracking platform for GNSS rovers/tablets from procurement to district allocation, surveyor handover, and repair.
- Built role-based authorization (Superadmin, DDLR, Vendor, Viewer) with automated audit logs, state alerts, and Recharts dashboards.

GeoServer User-Context Security Plugin | [Java](#), [GeoServer Plugin Architecture](#), [Spring Security](#)

- Developed a custom Java security plugin extending GeoServer to enforce dynamic, district-level layer filtering based on user context.

EDUCATION & PROJECTS

Education: B.E. in Information Science & Engineering — APS College of Engineering, Bengaluru (2018 – 2022)

Personal Projects: Cybersecurity Exam Platform (Node.js, Express, MongoDB, React) • Engineering Portfolio ([yogesh.is-a.dev](#), Next.js 15)